

Title: Lost in Translation: The Burden of Linguistic Marginalization in the Age of English-Centric AI Models in Malaysia

Abstract

This article investigates the sociolinguistic consequences of English-centric Artificial Intelligence (AI) models in multilingual Malaysia. As AI technologies integrate into daily life, their linguistic biases challenge non-Anglophone populations. Using a mixed-methods approach, this study examines AI performance disparities in English, Bahasa Malaysia, and Tamil. Findings reveal significant inadequacies in the models' handling of local languages, with higher error rates in key tasks. These technical shortcomings translate into tangible social and economic burdens, creating digital inequality and reinforcing English hegemony. We argue this technological linguistic marginalization functions as a modern mechanism of exclusion, limiting access for non-English speakers. The study concludes by calling for a de-centered, sociolinguistically informed approach to AI development that prioritizes linguistic equity.

Keywords: sociolinguistics, artificial intelligence, language models, Malaysia, digital inequality, linguistic marginalization

Introduction

The proliferation of Artificial Intelligence (AI) and Natural Language Processing (NLP) has fundamentally reshaped communication, commerce, and access to information globally. From virtual assistants and automated translation services to content moderation algorithms on social media, AI language models are increasingly embedded in the fabric of modern society. The development of these powerful tools, however, has been overwhelmingly dominated by data and research from English-speaking, Western contexts. The result is a technological ecosystem that is optimized for English, often at the expense of the world's vast linguistic diversity. This study explores the tangible consequences of this English-centricity in Malaysia, a nation characterized by its complex multilingualism, where Bahasa Malaysia (Malay), English, Mandarin, Tamil, and numerous indigenous languages coexist.

Malaysia's national language policy has historically navigated the tensions between promoting Malay as a language of national unity and recognizing the role of English in global commerce and higher education (Tsui & Tollefson, 2017). This sociolinguistic landscape provides a critical context for examining the impact of AI. When a technology designed primarily for one language is deployed in a multilingual society, who is left behind? This paper investigates what we term "technological linguistic marginalization": the process by which the design and implementation of technology create and reinforce social, economic, and cultural inequalities based on language.

We argue that the underperformance of AI models in Malaysian languages is not merely a technical inconvenience but a significant social burden. It creates barriers to digital literacy, skews the information ecosystem, and disadvantages non-English speakers in an increasingly digitized economy. By analyzing both the computational performance of AI systems and the lived experiences of Malaysian citizens, this research connects the micro-level frustrations of a mistranslated phrase or a misunderstood command to the macro-level structures of digital inequality and linguistic hegemony.

Literature Review

Sociolinguistics of a Multilingual Nation: Malaysia

Malaysia presents a compelling case study for the intersection of language, technology, and society. Its official language is Bahasa Malaysia, a status enshrined in the constitution to foster national identity. English, a legacy of the British colonial period, has been maintained as a powerful second language, indispensable in the domains of business, law, and tertiary education (Gill, 2013). This bilingual policy is further complicated by the widespread use of other languages, particularly Mandarin and Tamil, which are supported through a vernacular education system, and a rich diversity of indigenous languages.

Code-switching and the use of colloquial variants, such as "Manglish" (Malaysian English), are pervasive features of everyday communication (Thirusanku & Yunus, 2013). These fluid linguistic practices, while central to Malaysian identity, pose immense challenges for NLP models trained on standardized, monolingual datasets. Sociolinguistic research in Malaysia has extensively documented the ways in which language proficiency, particularly in English, correlates with socioeconomic status and access to opportunities, a concept Bourdieu (1991) defines as "linguistic capital." The introduction of English-centric AI threatens to create a new digital divide layered upon these existing inequalities, where technological proficiency becomes inseparable from English proficiency.

The Hegemony of English in AI and NLP

The field of NLP has made extraordinary advances, driven by the development of Large Language Models (LLMs) like GPT-4 and BERT (Devlin et al., 2019). These models are trained on vast quantities of text data, the majority of which is scraped from the internet. Given that over 60% of web content is in English, these models inevitably develop a deep and nuanced "understanding" of English, while their capabilities in other languages lag significantly behind (Bender et al., 2021).

Languages with fewer digital resources are termed "low-resource languages" in the NLP community. While Malay is better resourced than many languages, it still pales in comparison to English. Tamil and indigenous Malaysian languages are in an even more

precarious position. The lack of high-quality, large-scale, and digitally accessible corpora for these languages means that models struggle with their grammar, vocabulary, and cultural nuances (Joshi et al., 2020). This disparity is not accidental but a reflection of global economic and colonial histories that have privileged English as the lingua franca of science and technology. The result is a cycle of exclusion: languages with less economic power receive less investment in technological development, which in turn diminishes their utility in the digital sphere, further reducing their perceived value.

AI Bias and Digital Inequality

The concept of bias in AI extends beyond demographic representation to encompass linguistic and cultural biases. An AI model that consistently fails to understand or serve speakers of a particular language is inherently biased. This can manifest in various ways: a job application screening tool may unfairly penalize resumes written in non-standard Malay; a social media content moderation algorithm may fail to detect hate speech in Tamil; or a government e-service chatbot may be inaccessible to an elderly citizen who only speaks a local dialect.

These failures are not neutral. They actively create what could be called a "digital citizenship gap," where the ability to fully participate in an increasingly digital society is contingent upon proficiency in a dominant language (Hargittai, 2001). This exacerbates existing social inequalities, placing a disproportionate burden on communities that are already marginalized. The promise of AI to democratize access to information is thus inverted, becoming a tool that reinforces the very power structures it was meant to overcome.

Methodology

This study employed a sequential explanatory mixed-methods design to provide a comprehensive analysis of technological linguistic marginalization in Malaysia. This approach allowed us to first identify and quantify the performance disparities of AI models and then explore the qualitative, experiential dimensions of these disparities.

Phase 1: Computational Performance Analysis

In the first phase, we evaluated the performance of three widely accessible, pre-trained AI language models: Google Translate, OpenAI's GPT-3.5, and a popular sentiment analysis tool based on BERT architecture. We tested their capabilities in three core NLP tasks using English, standard Bahasa Malaysia, and standard Tamil.

1. **Machine Translation:** A dataset of 500 sentences was compiled, comprising common conversational phrases, news headlines, and government service announcements. These sentences were professionally translated from English into Malay and Tamil to create a parallel corpus. We then used the AI models to

translate the Malay and Tamil sentences back into English and measured the accuracy using the Bilingual Evaluation Understudy (BLEU) score, a standard metric for comparing machine-generated text to a set of high-quality human references.

2. **Sentiment Analysis:** We created a dataset of 300 social media comments (100 in each language) related to public services. Each comment was manually labeled as positive, negative, or neutral by three native speakers. We then ran the comments through the sentiment analysis tool and calculated the accuracy, precision, and recall for each language.
3. **Prompt Comprehension (for GPT-3.5):** We developed a set of 50 instructional prompts, such as "Summarize the following news article in three bullet points" or "Write a formal email complaining about poor internet service." The prompts were executed in all three languages, and the quality of the output was evaluated by native speakers on a 5-point Likert scale for relevance, coherence, and accuracy.

Phase 2: Qualitative Interviews

In the second phase, we conducted 25 semi-structured interviews to understand the lived experiences behind the quantitative data. Participants were recruited through purposive sampling to ensure a diverse range of perspectives and were divided into three groups:

1. **Everyday Users (n=15):** A mix of students, office workers, and retirees from diverse linguistic backgrounds (Malay, Chinese, and Indian ethnicities) who interact with AI-driven services regularly.
2. **Technology Developers (n=5):** Software engineers and product managers working in Malaysian tech companies.
3. **Policy Influencers (n=5):** Academics and civil servants involved in digital policy and language planning.

Interviews were conducted in the participant's preferred language (English, Malay, or Tamil) and lasted approximately 60 minutes. The discussions focused on their perceptions of AI, experiences with language barriers in technology, the challenges of developing non-English AI, and the potential societal impact of linguistic bias in technology. The interviews were transcribed, translated into English where necessary, and analyzed using thematic analysis to identify recurring patterns and themes.

Findings

The findings from both the computational analysis and the qualitative interviews converge to paint a clear picture of linguistic marginalization in Malaysia’s digital sphere.

Computational Disparities

The quantitative analysis revealed a stark hierarchy of AI performance, with English consistently outperforming both Bahasa Malaysia and Tamil across all tasks.

- **Machine Translation:** The BLEU scores for translation into English showed a significant drop-off for local languages. While the English-to-English baseline (translating professionally translated text back to the original) is not a standard measure, it serves to illustrate the models' internal consistency. The model achieved a high score for professionally translated English sentences. However, the score for Bahasa Malaysia was substantially lower, and for Tamil, it was lower still. Table 1 summarizes these results.

○ Table 1: Mean BLEU Scores for Machine Translation

Source Language	Mean BLEU Score
English (Baseline)	0.85
Bahasa Malaysia	0.62
Tamil	0.48

- Errors in Malay often involved incorrect idioms or overly formal phrasing, while errors in Tamil were more severe, frequently resulting in grammatically incoherent sentences. For example, the Malay phrase “*kerajaan akan kaji semula dasar itu*” (the government will review the policy) was sometimes translated as “government will study back that basic,” a literal but unnatural rendering.
- **Sentiment Analysis:** The sentiment analysis tool was significantly less accurate for Malay and Tamil comments compared to English. The model particularly struggled with sarcasm, code-switching, and culturally specific expressions. The accuracy for English was 88%, while it dropped to 65% for Malay and a mere 54% for Tamil. Negative comments in Malay and Tamil were frequently misclassified as neutral, a dangerous error for content moderation systems designed to detect harmful speech.
- **Prompt Comprehension:** The evaluation of GPT-3.5’s output revealed similar disparities. On the 5-point Likert scale, the average quality score for outputs from English prompts was 4.5. This dropped to 3.2 for Malay and 2.4 for Tamil. The

model often failed to follow complex instructions in Malay and Tamil, defaulting to generic or irrelevant responses. A prompt asking for a formal complaint email in Malay resulted in an overly simplistic text that lacked the appropriate honorifics and formal tone required in Malaysian official correspondence.

The Lived Experience of Marginalization

The qualitative interviews illuminated the human cost of these technological shortcomings. Three major themes emerged from the analysis: (1) The Burden of Adaptation, (2) Economic Disadvantage and Inequity, and (3) Erosion of Linguistic Confidence.

1. The Burden of Adaptation

A consistent theme among everyday users was the constant effort required to work around the linguistic limitations of technology. Rather than technology serving them, they felt they had to adapt their language to serve the technology. A university student, who primarily speaks Malay, explained:

"When I use the AI for my assignments, I can't just write in my normal Malay. I have to simplify my sentences, use very standard words. Sometimes I just give up and write my question in simple English, then translate the English answer back to Malay myself. It's extra work, a big hassle." (User 7, Malay speaker)

This "burden of adaptation" places a cognitive and temporal tax on non-English speakers. They are forced to become amateur translators and editors, a task not demanded of their English-speaking counterparts.

2. Economic Disadvantage and Inequity

For small business owners and professionals, the failures of AI had direct economic consequences. A small e-commerce entrepreneur who markets her products in both English and Malay described her frustration:

"I tried using an AI tool to write product descriptions. The English ones were great, very professional. The Malay ones were terrible, very stiff, sometimes the meaning was wrong. Customers will not trust a seller who cannot even write proper Malay. It makes my business look unprofessional. In the end, I have to write it all myself, which takes time away from other things." (User 11, Malay speaker)

A developer working for a local tech startup highlighted the systemic challenge. Building high-quality Malay or Tamil NLP features is expensive and time-consuming due to the lack of data and pre-existing tools.

"We want to build a better Malay chatbot for our customer service. But the open-source models are not good enough. To build one from scratch requires a massive dataset that

we don't have. The big tech giants have the data, but they are focused on the global market, which means English. We are left to pick up the scraps." (Developer 3)

This creates a vicious cycle where local companies cannot compete with global tech giants, reinforcing the dominance of English-centric platforms and limiting the growth of a localized digital economy.

3. Erosion of Linguistic Confidence

Perhaps the most subtle but insidious impact was the erosion of linguistic confidence. When technology consistently signals that one's native language is inferior or "unsupported," it can internalize a sense of linguistic insecurity. A policy influencer with a background in language advocacy noted:

"When a young person sees that all the 'cool' new technology works best in English, what message does that send? It says that their own language, Bahasa, is not modern, not important. It subtly pushes them towards English, not out of choice, but out of necessity. This is a digital form of linguistic assimilation." (Policy Influencer 2)

This sentiment was echoed by users who felt a sense of embarrassment when a voice assistant failed to understand their Malay commands, leading them to default to English in public settings to avoid the friction. The technology, in effect, becomes an agent of language shift, devaluing local languages in the high-status domain of modern technology.

Discussion

The findings of this study demonstrate that the English-centricity of AI models is not a neutral technical issue but an active generator of social inequality in Malaysia. The quantitative data provides empirical evidence of a linguistic performance gap, while the qualitative data reveals how this gap translates into a "burden of translation" and a "tax of adaptation" for non-English speakers. This reality directly challenges the utopian narrative of AI as a great equalizer. Instead, in the Malaysian context, AI appears to be amplifying the existing linguistic capital associated with English proficiency.

The struggles of users and developers reflect the dynamics of the global digital economy. Local innovators are at a structural disadvantage, unable to compete with the data-rich, resource-heavy ecosystems of Silicon Valley. This creates a dependency on technologies that are not designed with the Malaysian sociolinguistic context in mind. The failure of AI to comprehend code-switching, local idioms, and formal honorifics is a failure to recognize the linguistic reality of Malaysia. The models are, in essence, digitally tone-deaf, imposing a standardized, monolingual logic onto a fluid, multilingual society.

This technological marginalization has profound implications for digital citizenship. As government services and civic discourse move online, the inability of AI-driven

platforms to function equitably in local languages threatens to disenfranchise large segments of the population. An elderly citizen unable to use an e-government portal because its chatbot does not understand their dialect is as disenfranchised as one who cannot access a physical government office. The burden this places on individuals reinforces a digital divide that is not just about access to devices, but about the very language of access.

The erosion of linguistic confidence observed in this study is perhaps the most concerning long-term consequence. If technology implicitly teaches the next generation that innovation speaks only English, it could accelerate language shift and endanger the vitality of local languages. This runs counter to Malaysia's national policies aimed at promoting Bahasa Malaysia and preserving linguistic diversity. Technology, therefore, becomes a de facto language policy shaper, operating outside the deliberate, democratic processes of the state.

Conclusion

This study has provided a multi-faceted account of technological linguistic marginalization in Malaysia. We have shown that the technical biases of English-centric AI models impose real and significant social and economic burdens on Malaysian citizens. These burdens include the additional labor of adapting one's language for technology, exclusion from economic opportunities, and the subtle erosion of confidence in local languages. The digital landscape, far from being a level playing field, is becoming another arena where the hegemony of English is asserted, and linguistic diversity is penalized.

The limitations of this study include its focus on only three languages and a limited set of AI tasks. Future research should expand this analysis to include Mandarin, indigenous languages, and other colloquial variants like Manglish. Further investigation is also needed into specific domains such as healthcare and education, where the consequences of linguistic AI bias can be particularly severe.

Addressing this challenge requires a concerted effort from multiple stakeholders. Tech companies must move beyond a one-size-fits-all approach and invest seriously in developing models that are sociolinguistically aware and locally attuned. This involves not just collecting more data, but working with local communities to understand linguistic nuances and cultural contexts. Policymakers have a crucial role to play in creating standards for linguistic equity in technology, funding local AI research, and supporting the creation of high-quality digital resources for local languages. Educators and language advocates must continue to champion the value of multilingualism, ensuring it is seen not as a barrier to technological progress, but as an asset to be cultivated and preserved.

Ultimately, the goal must be to build an AI ecosystem that reflects the world's linguistic diversity, rather than one that flattens it. If AI is to fulfill its promise of empowering humanity, it must first learn to speak our languages.

References

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021, March). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency* (pp. 610-623).

Bourdieu, P. (1991). *Language and symbolic power*. Harvard university press.

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019, June). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)* (pp. 4171-4186).

Gill, S. K. (2013). *Language policy challenges in multi-ethnic Malaysia* (Vol. 8). Springer Science & Business Media.

Hargittai, E. (2001). Second-level digital divide: Mapping differences in people's online skills. *arXiv preprint cs/0109068*.

Joshi, P., Santy, S., Budhiraja, A., Bali, K., & Choudhury, M. (2020). The state and fate of linguistic diversity and inclusion in the NLP world. *arXiv preprint arXiv:2004.09095*.

Thirusanku, J., & Yunus, M. M. (2013). Malaysian English—a distinct variety. *Internet Journal of Language, Culture and Society*, 37, 18-29.

Tsui, A. B., & Tollefson, J. W. (Eds.). (2017). *Language policy, culture, and identity in Asian contexts*. Routledge.